

Week 6 Tutorial

Estimating a Firm's Cost of Capital

Zirui Song
April 2025

Agenda

- 4-1
- 4-2
- 4-3
- 4-4
- 4-5
- 4-6
- 4-8

Knowledge review 1: WACC

Figure 4-1 Matching Cash Flows and Discount Rates in DCF Analysis

Steps	Object of the Valuation Exercise	
	Equity Valuation	Project or Firm Valuation (Debt Plus Equity Claims)
Step 1: Estimate the amount and timing of future cash flows (covered in Chapter 2).	Equity free cash flow	Project (firm) free cash flow
Step 2: Estimate a risk-appropriate discount rate (covered in Chapters 4 and 5).	Equity required rate of return	Combine debt and equity discount rates (weighted average cost of capital—WACC).
Step 3: Discount the cash flows (covered in Chapter 2).	Calculate the present value of the estimated Equity FCFs using the equity discount rate to estimate the value of the equity invested in the project.	Calculate the present value of the Project or Firm FCF using the WACC to estimate the value of the project (firm) as an entity.

Knowledge review 1: WACC (cont.)

Equation (4.1) defines WACC as the average of the estimated required rates of return for the firm's interest-bearing debt (k_d), preferred stock (k_p), and common equity (k_e). The weight used for each source of funds is equal to the proportions in which funds are raised. That is, w_d is the weight attached to debt, w_p is the weight associated with preferred stock, and w_e is the weight attached to common equity.

$$\text{WACC} = k_d(1 - T)w_d + \boxed{k_p w_p} + k_e w_e \quad (4.1)$$

The connection of WACC to the DCF estimate of firm value is captured in Equation (4.2):¹

$$\text{Firm Value}_0 = \sum_{t=1}^N \frac{E(\text{Firm FCF}_t)}{(1 + \text{WACC})^t} \quad (4.2)$$

Problem 1

4-1 THOUGHT QUESTION As a summer intern, you are asked to prepare a spreadsheet calculating the project free cash flow associated with a project your employer is considering. Initially your boss assumes that no debt will be used to fund the project. During your presentation to the committee that evaluates projects, you learn that, in fact, the project will be financed with 25% debt. Determine whether the following statements are true or false, and explain your answer:

- You need to go back to your office and adjust the project's free cash flows to include the interest on the debt.
- You need to go back to your office and adjust the project cash flows to update the taxes paid due to the tax shield provided by taking on debt.
- Your cash flow model does not need to be updated because the financing of the project does not affect the free cash flow calculation.

Problem 1 (cont.)



- A reflection on problem 2-10 (c).

2-10 PROJECT VALUATION Glentech Manufacturing is considering the purchase of an automated parts handler for the assembly and test area of its Phoenix, Arizona, plant. The handler will cost \$250,000 to purchase plus \$10,000 for installation. If the company undertakes the investment, it will automate part of the semiconductor test area and reduce operating costs by \$70,000 per year for the next ten years. Five years into the life of the investment, however, Glentech will have to spend an additional \$100,000 to update and refurbish the handler. The investment in the handler will be depreciated using straight-line depreciation over ten years, and the refurbishing costs will be depreciated over the remaining five-year life of the handler (also using straight-line depreciation). In ten years, the handler is expected to be worth \$5,000, although its book value will be zero. Glentech's tax rate is 30%, and its opportunity cost of capital is 12%. Exhibit P2-10.1 contains cash flow calculations for the project that can be used in performing a DCF evaluation of its contribution to firm value. Answer each of the following questions concerning the project:

- Is this a good project for Glentech? Explain your answer.
- What can you tell about the project from the NPV profile found in Exhibit P2-10.1?
- If the project were partially financed by borrowing, how would this affect the investment cash flows? How would borrowing a portion of the investment outlay affect the value of the investment to the firm?
- The project calls for two investments: one immediately and one at the end of year 5. How much would Glentech earn on its investment? How should you account for the additional investment outlay in your calculations?
- What are the considerations that make this investment somewhat risky? How would you investigate the potential risks of this investment?

Figure 2-2 Calculation of Free Cash Flow (FCF)

Sales	Note that interest expense is <i>not</i> deducted before calculating the firm's tax liability. This is because FCF represents the cash flow available for both creditors (interest plus principal) and owners. In Chapter 9, where we calculate equity FCF, we will deduct interest expense before calculating the firm's tax liability.
Less: Cost of goods sold	
Gross Profit	
Less: Operating expenses (including depreciation expense)	
Earnings Before Interest and Taxes	
Less: Taxes	Note that from this point, the FCF calculation differs from the normal format of the income statement: (i) depreciation expense is added back to reflect the fact that although tax deductible, it does not represent a cash outflow; (ii) capital expenditures are deducted because they represent a cash outflow; and (iii) increases in net working capital are deducted because they, like capital expenditures for fixed assets, represent a cash outflow for an investment in the firm or project.
Equals: Net operating profit after taxes (NOPAT)	
Plus: Depreciation expense	
Less: Capital expenditures (CAPEX)	
Less: Increases in net working capital (Δ NWC)	
Free cash flow (FCF)	

Answer: (1) Despite changes in capital structure, the investment cash flows (FCF) remain **unchanged**. (2) However, increasing borrowing decreases WACC and increases NPV, thereby making the value of the investment (the project) **more attractive**.

Problem 1 (cont.)

Answer

- a. False, free cash flows are calculated independently of how a project is financed.
- b. False, The taxes used in FCF are based on EBIT (Earnings Before Interest and Taxes), not EBT (Earnings After Interest).
- c. True.

Knowledge review 2: YTM

$$\text{WACC} = k_d(1 - T)w_d + k_pw_p + k_e w_e$$

To elaborate on the distinction between promised yields and expected returns on debt, consider the calculation of YTM on a bond with one year to maturity in Equation (4.3a):

$$\frac{\text{Bond Price}}{\text{Price}} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})} \quad (4.3a)$$

Note that the cash flows used to determine YTM are the contractual or promised cash payments, which equal expected cash flows *only* for default-free bonds. For debt with default risk, the expected cash flows must reflect the probability of default (Pb) and the recovery rate (Re) on the debt in the event of default. For our one-period bond issue, the cost of debt k_d can be calculated using expected cash flows to the bondholder as follows:

Bond Price

$$= \frac{[\text{Interest} + \text{Principal}] \times (1 - Pb) + [\text{Interest} + \text{Principal}] \times Pb \times Re}{(1 + k_d)} \quad (4.3b)$$

Knowledge review 2: YTM (cont.)

Promised vs. expected

- The yield to maturity (YTM) is calculated using the promised interest and principal payments only when the **risk of default** is so low that promise cash flow are reasonable estimate of expected cash flow.
- For low-rated debt, however, promised and expected cash flows are **not the same**, and an explicit adjustment for the prospect of loss in the event of default becomes necessary.

Figure 4-2 A Guide to Corporate Bond Ratings

Moody's	S&P	Fitch	Definitions
Aaa	AAA	AAA	Prime (maximum safety)
Aa1	AA+	AA+	High grade, high quality
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Noninvestment grade
Ba2	BB	BB	Speculative
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC	—	In poor standing
Caa3	CCC-	—	
Ca	—	—	Extremely speculative
C	—	—	May be in default
—	—	DDD	Default
—	—	DD	
—	D	D	

$$YTM_{\text{promised}} = YTM_{\text{expected}}$$

Investment grade bond

Lower-rate bond

$$YTM_{\text{promised}} > YTM_{\text{expected}}$$

Knowledge review 2: YTM (cont.)

The expected cash flow to the bondholder is the promised principal and interest payments, weighted by the probability that the debt does not default ($1 - Pb$), plus the cash flow that is received in the event of default, weighted by the probability of default (Pb). To illustrate this, consider a bond with a \$1,000 face value and one year to maturity that is currently selling for \$985.00 and paying a 9% interest payment annually. Using Equation (4.3a), we estimate the bond's YTM to be 10.66%, that is:

$$\frac{\text{Bond Price}}{1} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})^1} = \frac{.09 \times \$1,000 + \$1,000}{(1 + .1066)} = \$985$$

$\rightarrow YTM_{\text{promised}} = 10.66\%$

The 10.66% YTM represents the rate of return an investor would realize if the bond does not default. If the probability of default on the bond is 15% and the recovery rate is 75%, however, the expected rate of return to the investor, using Equation (4.3b), is 6.51%.

$$\frac{(.09 \times 1000 + 1000) \times (1 - 0.15 + 0.15 \times 0.75)}{(1 + YTM_{\text{expected}})} = 985, YTM_{\text{expected}} = 6.51\%$$

RULE: IF $CF_{\text{promised}} > CF_{\text{expected}}$
 $YTM_{\text{promised}} > YTM_{\text{expected}}$

Problem 2

4-2 THOUGHT QUESTION Describe the difference between a promised cash flow and an expected cash flow. If promised cash flows tend to be higher than expected cash flows, should they be discounted at rates that are higher or lower than the firm's WACC?

Problem 2 (cont.)

RULE: IF $CF_{promised} > CF_{expected}$
 $YTM_{promised} > YTM_{expected}$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_e w_e \quad (4.1)$$

$$YTM_{promised}(1 - T)w_d + k_pw_p + k_e w_e > YTM_{expected}(1 - T)w_d + k_pw_p + k_e w_e$$

Discount rate > WACC

Answer

- Promised cash flows generally ignore some negative event, like project failure or political risk, and as a result, should be discounted at a higher rate.

Problem 3

4-3 CALCULATING A FIRM'S WACC Nestlé Enterprises is estimating its cost of capital for the first time and has made the following estimates: The firm's debt carries a AAA rating, which is currently yielding 6%; the firm pays taxes at a rate of 30%; the cost of equity is estimated to be 14%; and the firm's debt is equal to 20% of its enterprise value.

- a. What is Nestlé's estimated WACC?
- b. If Nestlé were to increase its debt level to 40% of enterprise value, the firm's investment banker has told the firm that its credit rating would drop to AA and correspondingly its cost of debt financing would rise to 7%. If the cost of equity corresponding to this new capital structure were to rise to 16%, what would be the firm's estimated WACC?

Problem 3 (cont.)

$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e \quad (4.1)$$

Answer

- a.* $WACC = 6\% \times (1 - 30\%) \times 20\% + 14\% \times 80\% = 0.84\% + 11.20\% = 12.04\%$
- b.* $WACC = 7\% \times (1 - 30\%) \times 40\% + 14\% \times 60\% = 11.56\%$

Problem 4

4-4 CALCULATING THE PROMISED YTM In December of 2005, the Eastman Kodak Corporation (EK) had a straight bond issue outstanding that was due in eight years. The bonds are selling for 108.126%, or \$1,081.26, per bond and pay a semiannual interest payment based on a 7.25% (annual) coupon rate of interest. Assume that the bonds remain outstanding until maturity and that the company makes all promised interest and principal payments in a timely basis. What is the YTM to maturity to the bondholders in December of 2005?

Problem 4 (cont.)

$$\frac{\text{Bond Price}}{1 + \text{YTM}} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})} \quad (4.3a)$$

Answer

- $1,081.26 = \frac{1000 \times \frac{7.25\%}{2}}{(1 + \text{YTM})^1} + \frac{1000 \times \frac{7.25\%}{2}}{(1 + \text{YTM})^2} + \dots + \frac{1000 \times \frac{7.25\%}{2}}{(1 + \text{YTM})^{16}} + \frac{1000}{(1 + \text{YTM})^{16}}, \text{ YTM}_{\text{semi}} = 2.98\%$
- Use *Rate* function in Excel to calculate it
- To convert the semi-annual YTM to its annual YTM equivalent we perform the following computation: $(1 + \text{YTM}_{\text{semi}})^2 - 1$
- $\text{YTM}_{\text{annual}} = (1 + .0298)^2 - 1 = 6.05\%$

Knowledge review 3: CAPM

$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e$$

- CAPM stands for the **Capital Asset Pricing Model**. It describes the relationship between risk and expected return for assets, especially stocks.

The CAPM can be expressed as shown in Equation (4.5), which relates the required expected return of an investment to systematic risk:

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \quad (4.5)$$

where

k_{rf} = the risk-free rate of interest

β_e = the beta, or systematic risk of the company's common equity, which is estimated from a regression of a stock's return minus the risk-free rate on a market return, such as the S&P 500 return, minus the risk-free rate

k_m = the expected return on the overall market portfolio comprised of all risky assets

$(k_m - k_{rf})$ = the expected market risk premium (the expected return on the overall market minus the risk-free rate)

Knowledge review 3: CAPM (cont.)

AutoSave On Data to Estimate Beta & MRP No Label • Saved Search

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Nitro PDF Pro Acrobat

Clipboard Font Alignment Number Styles Cells Editing Analyze Data Sensitivity Add-ins Create PDF and Share link Share via Outlook Adobe Acrobat

H22 Calibri 11 A⁺ A⁻

B B I U Font

	B	C	D	E
11 '2022	4.			
12 '2022	4.			
13 '2022	4.5	6,807.30	-0.0198	0.002559
14 '2022	4.59	6,789.90	0.017583	-0.00024
15 '2022	4.51	6,791.50	-0.01976	-0.00041
16 '2022	4.6	6,794.30	-0.00217	0.005179
17 '2022	4.61	6,759.20	0.026375	0.016348
18 '2022	4.49	6,649.60	-0.00222	-0.00562
19 '2022	4.5	6,687.10	0.020203	0.012263
20 '2022	4.41	6,605.60	0	-0.00679
21 '2022	4.41	6,650.60	-0.00227	0.00437
22 '2022	4.42	6,621.60	0.041576	0.002313
23 '2022	4.24	6,606.30	-0.03477	0.000621
24 '2022	4.39	6,602.20	-0.01133	-0.01142
25 '2022	4.44	6,678.00	-0.00225	0.004502

Standard Error 0.017109454
Observations 252

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.036965	0.036965	126.2746	5.5E-24
Residual	250	0.073183	0.000293		
Total	251	0.110148			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 99.0%	Upper 99.0%
Intercept	0.000370865	0.001078	0.343927	0.73119	-0.00175	0.002495	-0.00243	0.00317
X Variable 1	1.341156645	0.11935	11.2372	5.5E-24	1.106098	1.576216	1.031368	1.650945

Beta

CLEAN QAN RIO S&P_ASX 200 Rf

Ready Accessibility: Good to go Display Settings 100%

My Portfolio

News

Markets

Research

Personal Finance

Videos

Streaming Now

Summary

News

Chart

Conversations

Statistics

Historical Data

Profile

Financials

Analysis

Options

Holders

Sustainability

NasdaqGS - Nasdaq Real Time Price • USD

Apple Inc. (AAPL)

☆ Follow

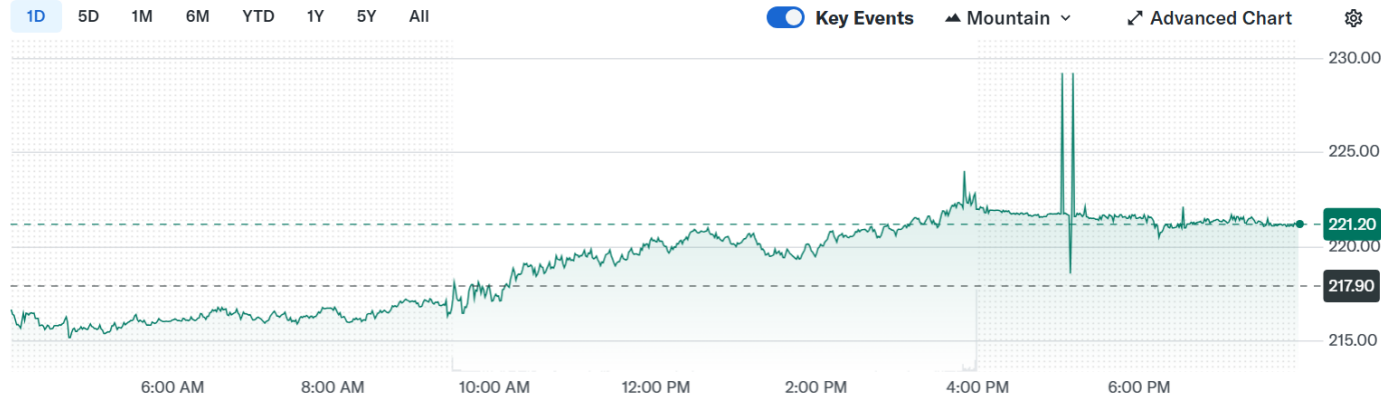
↔ Compare

🔔 Time to buy AAPL?

222.13 **+4.23** **+(1.94%)** **221.20** **-0.93** **(-0.42%)**

At close: 4:00:01 PM EDT

After hours: 7:59:58 PM EDT

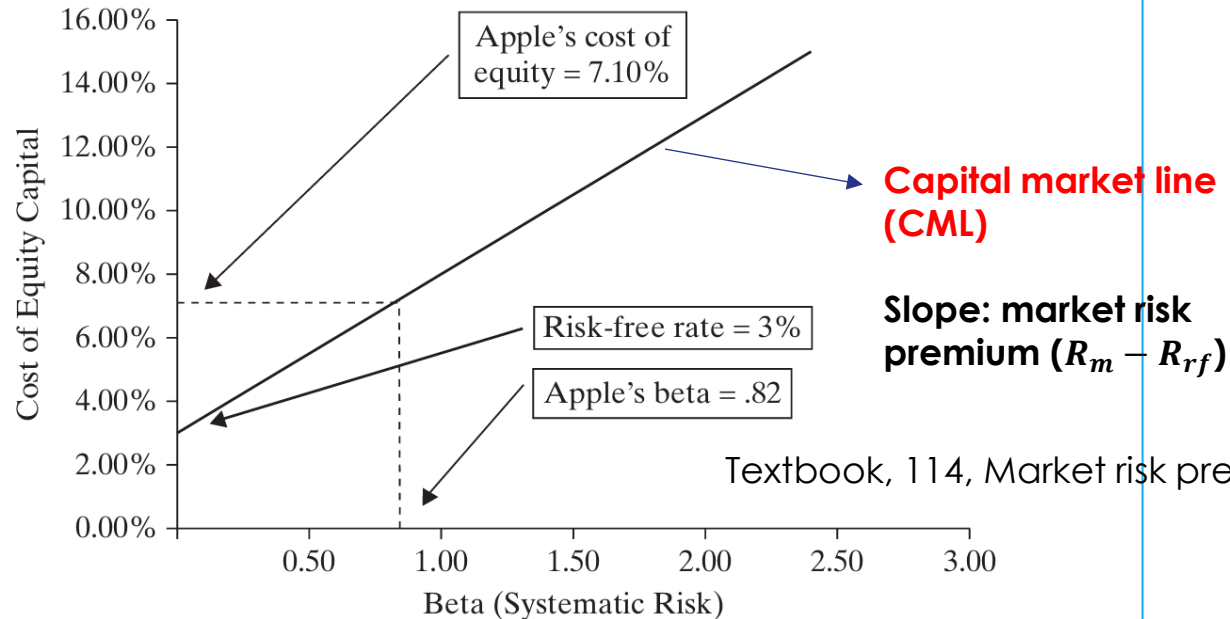


Previous Close	217.90	Day's Range	216.24 - 224.97	Market Cap (intraday)	3.337T	Earnings ...	Apr 30, 2025 - May 5, 2025
Open	217.22	52 Week Range	164.08 - 260.10	Beta (5Y Monthly)	1.18	Forward Dividend & Yield	1.00 (0.45%)
Bid	219.54 x 3000	Volume	65,078,252	PE Ratio (TTM)	35.20	Ex-Dividend Date	Feb 10, 2025
Ask	224.89 x 500	Avg. Volume	52,962,733	EPS (TTM)	6.31	1y Target Est	252.59

Knowledge review 3: CAPM (cont.)

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \quad (4.5)$$

Figure 4-4 Estimating Apple's Cost of Equity Using CAPM

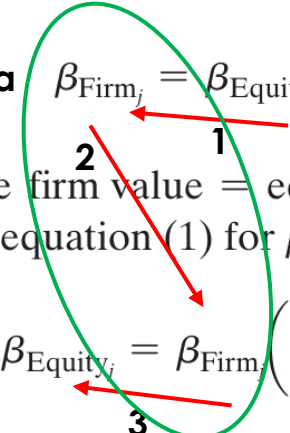


Knowledge review 3: CAPM (cont.)

When using the CAPM to estimate the cost of equity, it is standard practice to estimate the firm's equity beta by computing averages of equity betas of comparable firms with similar operating risk characteristics. Because equity betas vary with the use of financial leverage, we need to adjust the betas of the comparison firms for differences in how the firms are financed. This requires an understanding of the relationship between financial leverage and beta.

To develop an understanding of this relationship, we begin by noting that the beta for a firm is simply a weighted average of the betas of each of the sources of capital in the firm's capital structure. So the beta for firm j can be defined as follows:

Unlevered beta

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right) \quad (1)$$


Because firm value = equity value + debt value, we can solve for the equity beta by solving equation (1) for β_{Equity_j} as follows:

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) \quad (2)$$

Problem 5

4-5 CALCULATING THE COST OF EQUITY Caliber's Burgers and Fries is a rapidly expanding chain of fast-food restaurants, and the firm's management wants to estimate the cost of equity for the firm. As a first approximation, the firm plans to use the beta for McDonalds Corporation (MCD), which equals .56, as a proxy for its beta. In addition, Caliber's financial analyst looked up the current yield on ten-year US Treasury bonds and found that it was 4.2%. The final piece of information needed to estimate the cost of equity using the capital asset pricing model is the market risk premium, which is estimated to be 5%.

- Estimate the cost of equity for Caliber's using the CAPM and McDonalds Corporation's beta.
- McDonalds Corporation has an enterprise value of about \$80 billion and a debt of \$15 billion. If Caliber's has no debt financing, what is your estimate of Caliber's beta coefficient? You can assume that McDonalds' debt has a beta of .20.

Problem 5 (cont.)

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \quad (4.5)$$

Answer

a. $k_e = 4.2\% + .56 \times 5\% = 7\%$

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right) \quad (1)$$

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) \quad (2)$$

b. $\beta_{\text{Firm}} = 0.56 \times \frac{(80-15)}{80} + 0.2 \times \frac{15}{80} = 0.49$

Caliber's Levered equity beta: $\beta_{\text{Equity}} = 0.49 \times (1 + 0) - 0.2 \times 0 = 0.49$

Problem 6

4-6 CALCULATING THE COST OF EQUITY Smaltz Enterprises is currently involved in its annual review of the firm's cost of capital. Historically, the firm has relied on the CAPM to estimate its cost of equity capital. The firm estimates that its equity beta is 1.25, and the yield to maturity on long-term US Treasury bonds is 4.28%. The firm's CFO is currently in a debate with one of the firm's advisers at its investment bank about the level of the market risk premium. Historically, Smaltz has used 7% to approximate the market risk premium. However, the investment banker argues that this premium has shrunk dramatically in recent years and is more likely to be in the 3% to 4% range.

- a. Estimate Smaltz's cost of equity capital using a market risk premium of 3.5%.
- b. Smaltz's capital structure is comprised of 75% equity (based on current market prices) and 25% debt on which the firm pays a yield of 5.125% before taxes at 25%. What is the firm's WACC using both a 3% and 4% market risk premium?

Problem 6 (cont.)

Answer

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \quad (4.5)$$

a. $k_e = 4.28\% + 1.25 \times 3.5\% = 8.66\%$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e \quad (4.1)$$

b. $WACC = 5.125\% \times (1 - 25\%) \times 25\% + 8.66\% \times 75\% = 7.45\%$

Problem 7

4-8 THREE-STEP PROCESS FOR ESTIMATING A FIRM'S WACC Compano Inc. was founded in 1986 in Baytown, Texas. The firm provides oil-field services to the Texas Gulf Coast region, including the leasing of drilling barges. Its balance sheet for year-end 2014 describes a firm with \$830,541,000 in book value assets that has a market value of \$1.334 billion.

	December 31, 2014	
	Balance Sheet (Book Values)	Invested Capital (Market Values)
Liabilities and Owners' Capital		
Current liabilities		
Accounts payable	\$ 8,250,000	
Notes payable	—	—
Other current liabilities	7,266,000	
Total current liabilities	\$ 15,516,000	\$ —
Long-term debt (8.5% interest paid semiannually, due in 2015)	\$420,000,000	\$ 434,091,171
Total liabilities	\$435,516,000	\$ 434,091,171
Owners' capital		
Common stock (\$1 par value per share)	\$ 40,000,000	
Paid-in-capital	100,025,000	
Accumulated earnings	255,000,000	
Total owners' capital	\$395,025,000	\$ 900,000,000
Total liabilities and owners' capital	\$830,541,000	\$1,334,091,171

Compano's executive management team is concerned that its new investments be required to meet an appropriate cost of capital hurdle before capital is committed. Consequently, the firm's CFO has initiated a cost of capital study by one of his senior financial analysts, Jim Tipolli. Jim's first action was to contact the firm's investment banker to get input on current capital costs.

Jim learned that, although the firm's current debt capital required an 8.5% coupon rate of interest (with annual interest payments and no principal repayments until 2025), the current yield on similar debt would decline to 8% if the firm were to raise debt funds today. When he asked about the beta for Compano's debt, Jim was told that it was standard practice to assume a beta of .30 for the corporate debt of firms such as Compano.

- What are Compano's capital structure weights for debt and equity that should be used to compute its cost of capital?
- Based on Compano's corporate income tax rate of 40%, the firm's mix of debt and equity financing, and an unlevered beta estimate of .90, what is Compano's levered equity beta? (*Hint:* Compano plans on maintaining the mix of financing over time).
- Assuming a long-term US Treasury bond yield of 5.42% and an estimated market risk premium of 5%, what should Jim's estimate of Compano's cost of equity be if he uses CAPM?
- What is your estimate of Compano's WACC?

Problem 7 (cont.)

- a. We need to use the *market value* capital structure when calculating WACC.

$$\text{Debt value} = \frac{\text{Debt}}{\text{Enterprise value}} = \frac{434,091,171}{1,334,091,171} = 32.89\%$$

$$\text{Equity value} = \frac{\text{Equity}}{\text{Enterprise value}} = \frac{900,000,000}{1,334,091,171} = 67.11\%$$

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) \quad (2)$$

$$b. \beta_{\text{Equity}} = 0.9 \times \left(1 + \frac{32.89\%}{67.11\%} \right) - 0.3 \times \frac{32.89\%}{67.11\%} = 1.19$$

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \quad (4.5)$$

$$c. k_e = 5.42\% + 1.19 \times 5\% = 11.39\%$$

$$d. WACC = 8\% \times (1 - 40\%) \times 32.89\% + 11.39\% \times 67.11\% = 9.22\%$$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e \quad (4.1)$$